
Project Report - ECE 176

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Abstract

In this project, we explore self-supervised representation learning using context-based image inpainting, inspired by the paper “Context Encoders: Feature Learning by Inpainting.” The goal is to learn meaningful visual features without labeled data by training a neural network to predict missing regions of an image from its surrounding context.

We implement an encoder-decoder convolutional neural network that takes a corrupted image as input and reconstructs the missing region. To improve visual quality, we incorporate both reconstruction loss and adversarial loss. Our results demonstrate that the model is capable of generating semantically consistent image completions and learning useful feature representations for downstream tasks.

1 Introduction

Learning effective visual representations traditionally relies on large labeled datasets, which are expensive and time-consuming to obtain. To address this limitation, self-supervised learning methods aim to learn meaningful features directly from unlabeled data by designing surrogate tasks that provide supervisory signals.

In this project, we explore self-supervised representation learning through the task of image inpainting, based on the work of Pathak et al. (2016). The core idea is to remove a region of an image and train a model to predict the missing content using the surrounding context. Successfully completing this task requires the model to capture both low-level texture information and high-level semantic structure, such as object shapes and spatial relationships.

We implement a context encoder model using a convolutional encoder-decoder architecture. To improve reconstruction quality beyond the original formulation, we incorporate several modifications, including a U-Net style generator with skip connections, a patch-based discriminator that focuses on the missing region, and a combination of reconstruction and adversarial losses. We also include a total variation regularization term to encourage spatial smoothness in the generated outputs.

Our results demonstrate that combining contextual reconstruction with adversarial training leads to significantly improved inpainting performance. In particular, skip connections help preserve structural details, while adversarial loss enhances the realism of generated regions. Overall, this project highlights the effectiveness of inpainting as a self-supervised learning task and provides insight into how architectural and training choices impact model performance.

2 Related Work

Self-supervised learning has emerged as a powerful approach for learning visual representations without labeled data. Instead of relying on human annotations, these methods define pretext tasks that generate supervision directly from the data itself.

Early self-supervised approaches focused on pretext task-based methods, where models are trained to solve auxiliary tasks that require semantic understanding of images. For example, Noroozi and

Favaro (2016) proposed solving jigsaw puzzles as a way to learn spatial relationships between image patches, demonstrating that such tasks can produce useful representations for downstream vision problems. Another line of work includes predicting image rotations (RotNet) and cross-channel prediction, both of which encourage models to capture structural and semantic information.

Context encoders, which we implement in this project, also fall under this category. By predicting missing regions of an image, the model must infer both local textures and global scene structure, making it a strong self-supervised signal.

Another important class of methods is autoencoder-based approaches, which learn representations by reconstructing input data. While effective, standard autoencoders often produce blurry outputs and may fail to capture high-level semantics unless additional constraints are introduced. The context encoder improves upon this by combining reconstruction with adversarial learning.

More recent advances in self-supervised learning are dominated by contrastive learning methods. These approaches learn representations by maximizing agreement between different augmented views of the same image while distinguishing them from other images. A notable example is SimCLR (Chen et al., 2020), which demonstrates that contrastive learning can achieve performance competitive with fully supervised models on ImageNet. Other methods such as MoCo, BYOL, and SimSiam further improve representation quality through memory mechanisms, momentum encoders, or non-contrastive objectives.

Recent surveys categorize self-supervised learning methods into major groups including context-based, generative, and contrastive approaches, highlighting the evolution from hand-designed pretext tasks to large-scale representation learning frameworks.

The context encoder approach used in this project lies at the intersection of pretext task-based and generative methods. Unlike contrastive approaches that rely on instance discrimination, context encoders directly model image structure through reconstruction, making them particularly suitable for tasks such as image inpainting and semantic understanding.

Overall, self-supervised learning methods have significantly reduced the reliance on labeled datasets and have become a fundamental component of modern computer vision pipelines.

3 Method

Our implementation is based on the context encoder framework for image inpainting, with several modifications to improve reconstruction quality and training stability.

3.1 Problem Setup

Given an input image, we remove a central square region and train a model to predict the missing content using the surrounding pixels. Let x denote the original image and M be a binary mask where the missing region is set to zero. The input to the model is the masked image:

$$x_{\text{masked}} = x \odot M$$

The goal is to reconstruct the missing region while preserving consistency with the visible context.

3.2 Model Architecture

Generator. We use a U-Net style encoder-decoder architecture with skip connections. The encoder consists of a series of convolutional layers that progressively downsample the input and extract hierarchical features. The decoder upsamples these features to reconstruct the full image.

Skip connections between corresponding encoder and decoder layers allow the model to retain spatial details from the visible regions, which is critical for realistic inpainting.

The generator outputs a full image $\hat{x}$, which is combined with the known pixels:

$$x_{\text{completed}} = x \odot M + \hat{x} \odot (1 - M)$$

Discriminator. We use a patch-based discriminator that operates only on the missing region rather than the entire image. This focuses adversarial learning on the region that must be generated, improving realism without penalizing already correct pixels.

3.3 Loss Functions

We train the model using a weighted combination of reconstruction, adversarial, and regularization losses.

- **Reconstruction Loss:** We use an L_1 loss between the reconstructed and ground truth images, with higher emphasis on the missing region:

$$\mathcal{L}_{rec} = \|(1 - M) \odot (x - \hat{x})\|_1$$

where $\hat{x} = F(x \odot M)$ is the reconstructed image.

- **Adversarial Loss:** A patch discriminator is trained to distinguish real and generated patches. The generator is trained to fool the discriminator:

$$\mathcal{L}_{adv} = \mathbb{E}[\log D(x_{patch})] + \mathbb{E}[\log(1 - D(\hat{x}_{patch}))]$$

- **Total Variation Loss:** We include a total variation loss to encourage spatial smoothness:

$$\mathcal{L}_{tv} = \sum |\nabla \hat{x}_{patch}|$$

The final objective is:

$$\mathcal{L} = \lambda_{rec} \mathcal{L}_{rec} + \lambda_{adv} \mathcal{L}_{adv} + \lambda_{tv} \mathcal{L}_{tv}$$

3.4 Training Details

We train the model using the Adam optimizer with learning rate 2×10^{-4} . Images are resized to 128×128 , and a central 64×64 region is removed. Pixel values are normalized to $[-1, 1]$ to match the generator’s output activation.

During training:

- The generator predicts the full image from the masked input
- The completed image is formed by combining generated and known pixels
- The discriminator evaluates only the missing patch
- Generator and discriminator are updated alternately

4 Experiments

4.1 Dataset

We train our model on the CelebA dataset, which contains over 200,000 aligned face images. Each image is center-cropped and resized to 128×128 . Pixel values are normalized to the range $[-1, 1]$.

To construct the training task, a central 64×64 region is removed from each image, and the model is trained to reconstruct this missing region.

4.2 Training Setup

The model is trained using mini-batches of size 64. We use the Adam optimizer with learning rate 2×10^{-4} and momentum parameters $\beta_1 = 0.5$ and $\beta_2 = 0.999$.

Training is performed for multiple epochs, with sample reconstructions saved periodically to monitor progress. The generator and discriminator are trained jointly using adversarial learning.

4.3 Results

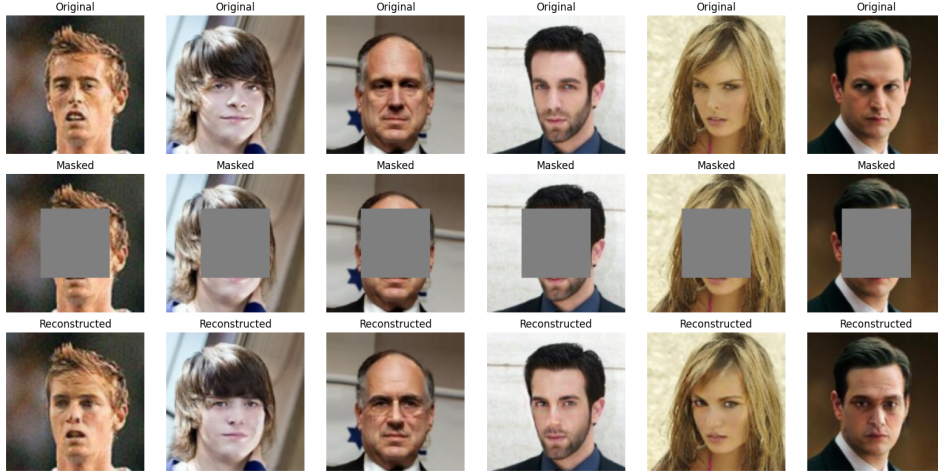


Figure 1: Image inpainting results. Top row: original images. Middle row: masked inputs. Bottom row: reconstructed outputs.

Figure 1 shows example results from the model. The first row contains the original images, the second row shows masked inputs, and the third row shows reconstructed outputs.

Compared to a baseline encoder-decoder model, our improved architecture produces:

- Sharper facial features
- Better structural consistency (eyes, nose, mouth alignment)
- Reduced blurring in the reconstructed region

The use of skip connections allows the model to better preserve spatial information, while adversarial training improves visual realism.

4.4 Ablation Insights

We observe that several design choices significantly improve performance:

- **Normalization to $[-1, 1]$:** Prevents mismatch between input and generator output, improving convergence.
- **Patch-based discriminator:** Focuses learning on the missing region, leading to more realistic completions.
- **U-Net architecture:** Skip connections help retain high-frequency details.
- **Weighted reconstruction loss:** Emphasizing the missing region improves reconstruction accuracy.

The use of a GAN specifically dramatically improves facial feature creation, making more realistic human-looking features.

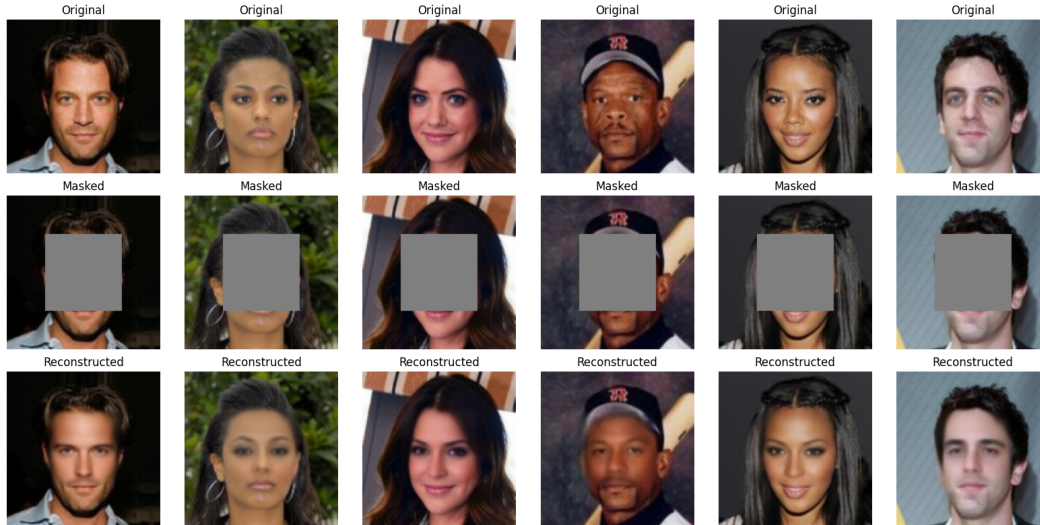


Figure 2: Image inpainting results when removing GAN

4.5 Limitations

Despite improvements, the model still struggles with:

- Highly ambiguous regions (e.g., occluded faces)
- Fine-grained details such as eyes and textures

These limitations arise from the inherent uncertainty of the inpainting task and the limited capacity of the model.

4.6 Future Work

Performance can be further improved by:

- Training for more epochs
- Using perceptual loss (e.g., VGG features)
- Incorporating random or irregular masks
- Increasing model capacity or resolution

References

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